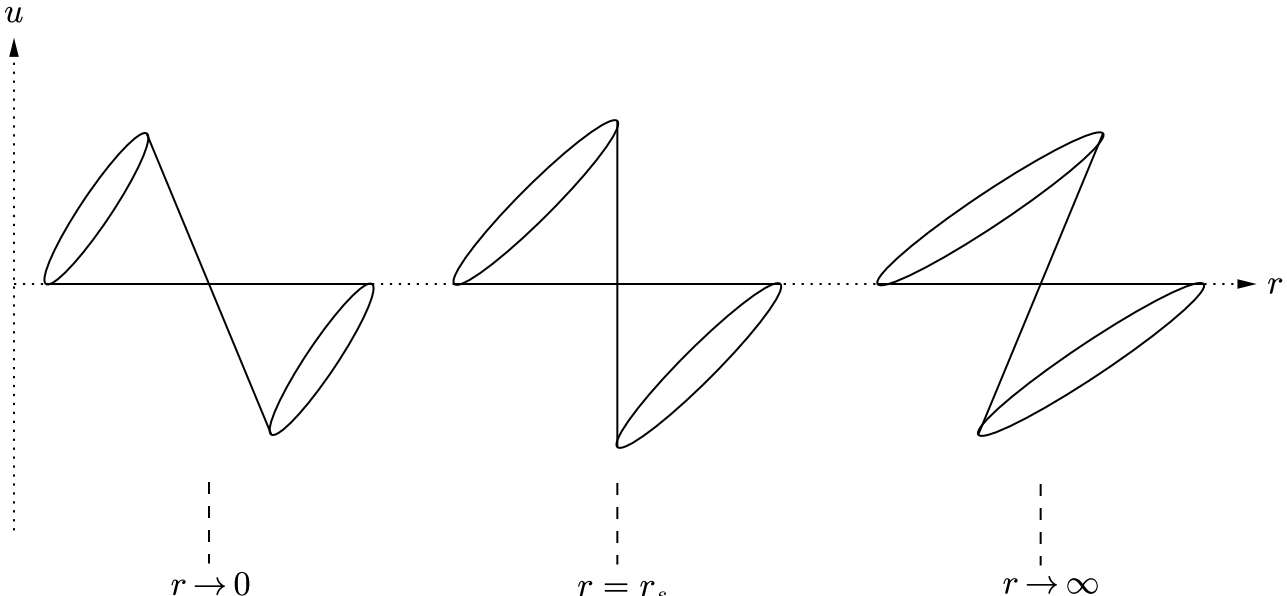
u
r
$r \to 0$
$r = r_s$
$r \to \infty$